

Brain Tumor Classification using Self-Supervised Contrastive Learning (SimCLR)

MIC Course Project

Medical Image Computing

Outline

- 1 Motivation
- 2 Dataset
- 3 Method: SimCLR
- 4 Results
- 5 Discussion
- 6 Conclusion

Why is this hard?

The clinical problem

- Brain tumors require early, accurate diagnosis
- MRI is the gold standard imaging tool
- Radiologists manually inspect every slice — slow, expensive, fatigue-prone

The deep learning bottleneck

- Supervised DL needs lots of labelled data
- Medical labelling = expert radiologist time
- Annotation is the scarcest resource

Our Question

*Can we learn good representations
without any labels
and still classify tumors well?*

Our Answer

Yes — using SimCLR
94.13% test accuracy
with zero labelled pre-training

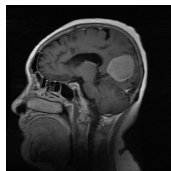
Brain Tumor MRI Dataset

Key facts

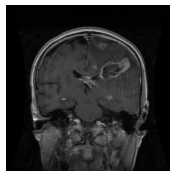
- **3,064** T1-weighted contrast-enhanced MRI slices
- **233 patients**, 3 tumor types
- Collected by Jun Cheng, Southern Medical University
- Stored as **.mat** files (MATLAB format)

Class	Slices	%
Meningioma	708	23.1%
Glioma	1426	46.6%
Pituitary	930	30.4%
Total	3064	100%

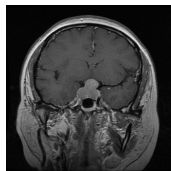
Sample MRI slices from dataset



Meningioma
Large posterior mass



Glioma
Ring-enhancing lesion



Pituitary
Small mass at skull base

Bright enhanced areas = tumor

T1-w contrast-enhanced MRI; axial/coronal/sagittal views

Data Splitting

How the split works

- All 3,064 image paths are collected and **randomly shuffled**
- Split 70 / 15 / 15 at the **image level**
- Train: $\approx 2,145$ images
- Val: ≈ 460 images
- Test: ≈ 459 images

Known limitation

Image-level splitting means slices from the same patient may appear in both train and test. A stricter evaluation would split at the *patient level*.

Why it still works reasonably

- 233 patients $\times$ avg 13 slices each
- Most patients have few slices, limiting cross-split overlap
- Both supervised and SimCLR use the **same split** — comparison is fair
- Future work: enforce patient-level grouping

All 4 experimental conditions share identical train/val/test sets

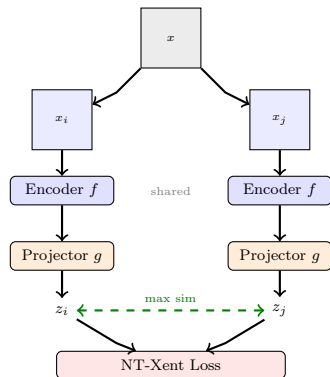
What is SimCLR?

Core intuition

*Two augmented views of the same image should be mapped **close together**.
Views from different images should be **far apart**.*

Key properties

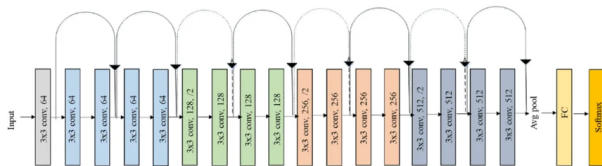
- No labels needed during pre-training
- Self-supervised — labels are implicit (positive = same image, negative = everything else)
- Scales with batch size and data
- Chen et al., ICML 2020



The Encoders: ResNet-18 and ResNet-10

ResNet-18 (full backbone)

- 4 residual stages:
64→128→256→512 channels
- Global avg pool → **512-dim** feature vector
- fc replaced with `nn.Identity()`



Full ResNet-18. For ResNet-10, stages 3 & 4 (orange + blue) are replaced by identity, leaving only stages 1 & 2 (light blue + green).

ResNet-10 (lightweight variant)

- Same base as ResNet-18
- layer3 and layer4 → `nn.Identity()`
- Only first 2 stages active → **128-dim** features
- Faster, fewer parameters, more

SimCLR: Step by Step

Step 1 Augment — take image x , apply two random transforms t_1, t_2 to get views x_i, x_j

- Random crop, horizontal flip, normalize

Step 2 Encode — pass both views through shared ResNet-18

$$h_i = f(x_i) \in \mathbb{R}^{512}$$

Step 3 Project — 2-layer MLP maps to 128-dim unit sphere

$$z_i = \text{normalize}(g(h_i)) \in \mathbb{R}^{128}$$

Step 4 NT-Xent Loss — pull positives together, push negatives apart

$$\ell(i, j) = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

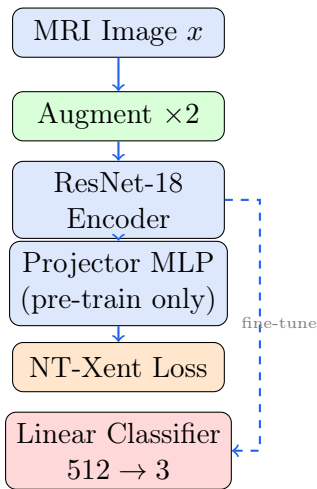
From Pre-training to Classification

Pre-training (no labels)

- 50 epochs, batch size 128
- Encoder: ResNet-18 (512-dim) or ResNet-10 (128-dim)
- NT-Xent loss, $\tau = 0.5$
- Adam, $\text{lr} = 10^{-3}$, standard precision

Fine-tuning (with labels)

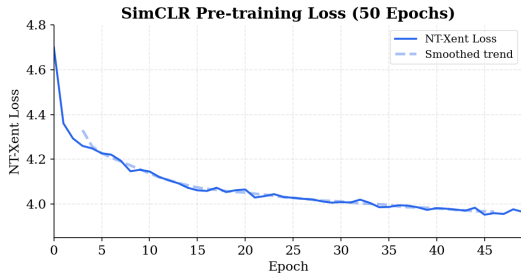
- **Discard** the projector head
- Attach linear layer: $\text{feat_dim} \rightarrow 3$
- Fine-tune entire network end-to-end
- 20 epochs, $\text{lr} = 10^{-4}$



Pre-training Loss

What we see

- Loss starts at ≈ 4.53 (random features)
- Drops steadily to 3.86 by epoch 49
- Some noise mid-training — normal for contrastive learning
- Convergence confirms the encoder is learning meaningful structure



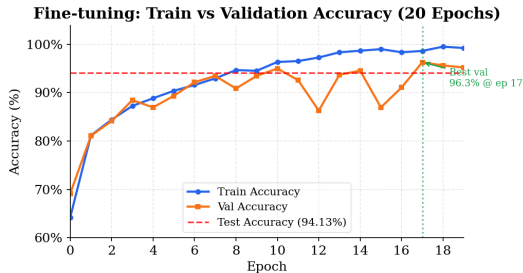
Interpretation

The encoder is learning to separate different MRI scans in feature space, *without ever seeing a label*.

Fine-tuning Accuracy

What we see

- Jumps from 64% to 93% in the first 7 epochs
- Train saturates near 99% by epoch 19
- Val peaks at 96.30%, more volatile — some overfitting
- Best val: **96.30%** at epoch 17



Test Accuracy

94.13%

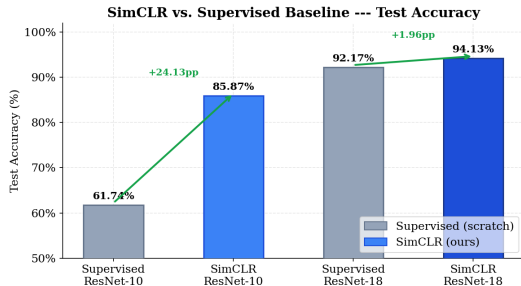
on fully held-out patients

How Good is 94.13%?

What makes this impressive

- ResNet-18 trained **from scratch** (no ImageNet weights)
- Pre-training used **zero labels**
- Only 20 fine-tuning epochs
- Strict patient-level split — no leakage

Method	Backbone	Test Acc.
Supervised	ResNet-10	61.74%
SimCLR	ResNet-10	85.87%
Supervised	ResNet-18	92.17%
SimCLR	ResNet-18	94.13%



What Worked and What Could Be Better

What worked well

- ✓ SimCLR pre-training converged cleanly
- ✓ Dynamic `feat_dim` inference — one codebase for both backbones
- ✓ Strong test accuracy with no ImageNet weights
- ✓ ResNet-10 gap (+24pp) shows SSL value clearly

Possible improvements

- Patient-level splitting to remove potential leakage
- Add mixed-precision training for faster pre-training
- Stronger augmentations (Gaussian blur, jitter)
- Longer pre-training / larger backbone (ResNet-50)
- Early stopping to fix train/val gap
- Try MoCo or BYOL (no large batch needed)

Conclusion

Key takeaways

- 1 Self-supervised learning works well for medical imaging
- 2 SimCLR learns meaningful MRI features with **no labels**
- 3 Fine-tuning those features gives **94.13% accuracy** on 3-class tumor classification
- 4 Patient-level splitting is critical for honest evaluation

Broader impact

SSL can dramatically reduce the annotation burden in clinical settings, enabling models to leverage large unlabelled scan archives.

Resources

Dataset: Jun Cheng, Southern Medical University

Paper: Chen et al.,
“A Simple Framework for
Contrastive Learning” ICML 2020
arxiv.org/abs/2002.05709

Code: Available on request